

WASP-11b

R-1559

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_211113 · created 2026-08-01T21:12:48+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459531.9062 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1576 +/- 0.0052
Transit depth (Rp/Rs)^2	2.49 +/- 0.16 [%]
Orbital Inclination (inc)	89.98 +/- 0.4
Airmass coefficient 1 (a1)	1.204 +/- 0.011
Airmass coefficient 2 (a2)	-0.173 +/- 0.004
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	None
Optimal Aperture	4.72
Optimal Annulus	17.23
Transit Duration (day)	0.1095 +/- 0.00065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1576 ± 0.0052 vs 0.131 (deviation 3.8σ)
Duration fit vs published	0.1095 d vs 0.1075 d (deviation 2.29σ)
Mid-time O–C	0.7 min
Depth SNR	22.5
Residual scatter	0.95 %

Light curve

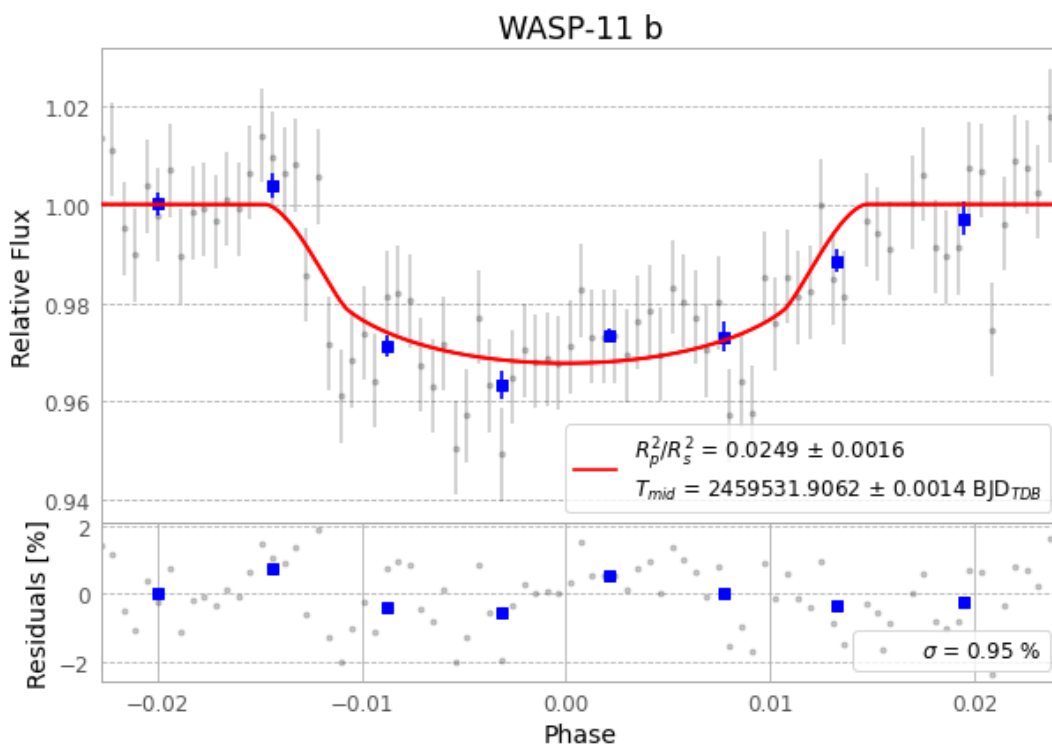


Figure 1 — FinalLightCurve_WASP-11 b_2021-11-13.png (SHA-256 6673ceaaf26ee3c6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7e12cbf1c80a1083...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-10211113073312.FITS → HATP-10211113114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-211113.log (06f13e14a19d...), FinalLightCurve_WASP-11 b_2021-11-13.png (6673ceaaf26e...), FinalLightCurve_WASP-11 b_2021-11-13.csv (2a21f6905822...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:12Z.